

Fengtao Wang

Waterloo, ON | ccbdchcc@gmail.com | 548-990-6600 | linkedin.com/in/fengtao-wang | github.com/asdfaqr

Education

University of Waterloo, Sept 2024 – June 2029
Candidate of **Bachelor of Mathematics** (Mathematical Finance and Economics)
Relevant Coursework: Computational Mathematics, Optimization, Differential Equations, Linear Algebra 1 & 2, Calculus 1–3, Probability & Statistics, Micro/Macroeconomic Theory

Awards & Achievements

- **University of Waterloo President's Scholarship of Distinction**
- **Euclid Mathematics Contest:** Certificate of Distinction (Top 25% nationally).
- **Fermat & Hypatia Mathematics Contests:** Honour Roll (Recognized for high standing).

Work Experience

Researcher – LiveSens, WeAccelerate (Data Driven Solutions) June 2025 – Aug 2025

- Analyzed multi-regional air quality datasets and sensor hardware specifications from market leaders (**Gaia, Envea, MetOne Instruments**) to synthesize environmental market trends.
- Built a **vanilla JS visualization suite** to explore air quality data use-cases, forming the strategic framework for **data monetization** and commercialization of LiveSens datasets.

University Economics Tutor, University of Waterloo Sept 2025 – Dec 2025

- Guided student athletes through difficult economics concepts (e.g., Indifference curves, Marginal costs, Equilibrium).

Projects

TroupeTTS (AI Audio Pipeline) - Typescript + NodeJS

troupetts.com | github.com/FengchiW/Troupe

- Engineered a pipeline to convert text/images into diarized audio using Gemini and Inworld APIs.
- Implemented automated speaker labeling and multi-modal language understanding for complex documents.

Discrete Autoencoder for Mathematical Reasoning - PyTorch + Python

github.com/asdfaqr/Math-Reasoning-Discrete-Autoencoder

- Built a two-half architecture: an encoder mapping mathematical reasoning steps into discrete bottleneck codebooks, paired with an autoregressive Transformer decoder for exact reconstruction.
- Outperformed standard lossless baselines with **37% higher compression than gzip**.

Chess CNN (Deep Learning) - Python + JS/HTML

github.com/asdfaqr/CNN-Chess-Project

- Architected a CNN using Residual/SE blocks trained on over 2 million positions; achieved **40+% accuracy** on Stockfish moves and **~1500 Elo** without search.
- Built a web-based suite to benchmark model capabilities.

Interactive Mind Map (Visualization) - JS/HTML

github.com/asdfaqr/mindmap_project

- Developed a dynamic and interactive mind map using vanilla JS to visualize data hierarchies for client presentations.
- Developed a dynamic documentation interface where selecting individual nodes retrieves comprehensive, pre-structured technical summaries within a dedicated side-panel.

Crazyhouse Bots (Algorithmic Search) - Python

github.com/asdfaqr/Crazyhouse_bots

- Developed an interactive UI in Python using PyGame, implementing event-driven logic for piece 'drops' and real-time valid-move highlighting to handle unique Crazyhouse mechanics.
- Implemented minimax with alpha-beta pruning and MCTS for a Crazyhouse variant.

Relevant Skills

Languages: C++, C, C#, Python, Java, JavaScript, HTML/CSS, SQL, R, Processing

Tools: PyTorch, NumPy, Git, Bash/Shell, Linux, Excel (Data Analysis), PowerPoint